

Experimental data for a liquid/vapor binary mixture of pentane (P) and heptane (H) is given in the table. The quantity z_{liq} gives the mole fraction of pentane in the liquid phase while z_{vap} gives the mole fraction of pentane in the vapor phase. The data can be linearized using the relation $Y = AX$ where A is the separation factor, $X = z_{liq}/(1 - z_{liq})$, and $Y = z_{vap}/(1 - z_{vap})$.

For a general straight line with the intercept equal to zero $y = ax$, the "best fit" slope for perpendicular offsets is given by $a = -\bar{B} \pm \sqrt{\bar{B}^2 + 1}$ where $\bar{B} = \frac{S_{xx} - S_{yy}}{2S_{xy}}$ and the

averages are defined as $S_{xx} = \frac{1}{n} \sum_{i=1}^n x_i^2$, $S_{yy} = \frac{1}{n} \sum_{i=1}^n y_i^2$, and $S_{xy} = \frac{1}{n} \sum_{i=1}^n x_i y_i$, respectively. If $S_{xx} - S_{yy} > 0$, use the plus sign in front of the square root in the expression for the slope. If $S_{xx} - S_{yy} < 0$, use the minus sign.

(a) Give the units on the mole fractions z_{liq} and z_{vap} .

(b) Give the units on the mole fraction ratios X and Y .

(c) Calculate the "best fit" value of the separation factor using perpendicular offsets as described above. Show all of your work.

Mole Fraction z_{liq}	Mole Fraction z_{vap}
0.867	0.984
0.594	0.925
0.398	0.836
0.254	0.701
0.145	0.521
0.059	0.271

(c)

X	Y	X^2	Y^2	XY
6.52	61.5	42.49	378.2	400.9
1.46	12.3	2.14	152	18.04
0.661	5.10	0.437	25.99	3.370
0.340	2.34	0.1154	5.497	0.7983
0.170	1.09	0.0287	1.183	0.1845
0.0623	0.372	0.0039	0.1382	0.02331

S_x	S_y	S_{xx}	S_{yy}	S_{xy}
1.536	13.791	7.537	661.2	70.55

$$\bar{B} = \frac{(7.537 - 661.2)}{2(70.55)} = -4.632$$

$$A = 4.632 \pm \sqrt{(-4.632)^2 + 1} = -0.1067 \quad (-) \quad \text{or} \quad 9.371 \quad (+)$$

(a) mol P/mol total
(b) mol P/mol H

$$a = -\bar{B} \pm \bar{B} \sqrt{1 + 1/\bar{B}^2}$$

$$(b) \frac{\frac{\text{mol P}}{\text{mol}}}{1 - \frac{\text{mol P}}{\text{mol}}} = \frac{\text{mol}}{\text{mol}}$$

$$= \frac{\text{mol P}}{\text{mol} - \text{mol P}} = \frac{\text{mol P}}{\text{mol H}}$$